Plan: 3-Orthogonal-Dimension Architectural Factorial
HSiKAN × CPML × Clifford-FIR

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1 Scope

We have spent the day disentangling architectural contributions by ablating one at a time. The pattern caught two false attributions already: "CPML breaks Epinions" was an embedding-lookup artefact, "+ricci-mod is best HyMeYOLO config" was a wrong-metric artefact. The pattern that keeps catching them is the *orthogonal-dimension factorial*: each architectural feature is toggled independently, every combination cell is measured, and attribution becomes a question of effect decomposition over the factorial design.

This plan formalises the **three architectural axes** that have emerged as the candidate ICLR / MLSys contributions of this codebase. Each is independently toggleable, and any subset can be active in a given model:

1. **HSiKAN** (Catmull-Rom signed-spline aggregator)
2. **CPML** (tier-stratified multi-layer topology)
3. **Clifford-FIR** (closed-form Clifford-derivative linear aggregator)

The factorial is $2^3 = 8$ cells, plus a baseline embedding-only floor for each cell (cycles on/off), giving 16 measurements per seed. With 5 seeds and 2 datasets (Bitcoin OTC + Epinions) this is 160 runs total. At observed warm-cache training cost (~ 30 s per Bitcoin OTC run, ~ 200 s per Epinions run) the full factorial costs ~ 1.5 hours of CPU time per dataset.

2 Mathematical decomposition

Let $A, C, F \in \{0, 1\}$ denote whether axes HSiKAN, CPML, FIR are active. Let L : dataset $\times$ $\{A, C, F\}^3 \to [0, 1]$ be the test AUC. The factorial decomposition is

$$L(D, A, C, F) = \mu_D + \alpha \cdot A + \gamma \cdot C + \varphi \cdot F + \alpha\gamma \cdot AC + \alpha\varphi \cdot AF + \gamma\varphi \cdot CF + \alpha\gamma\varphi \cdot ACF + \varepsilon,$$

where μ_D is the dataset's embedding-only floor and the six greek-letter coefficients are the (single-axis, pairwise, and three-way) effects. We solve for them by standard 2-level factorial regression on the 8-cell design.

Practically interesting null hypotheses:

- $\alpha = 0$: HSiKAN adds nothing (Catmull-Rom splines have no marginal effect over MLP/linear).
- $\varphi = 0$: Clifford-FIR adds nothing (closed-form Clifford derivative is no better than autograd MLP).
- $\alpha\varphi = 0$: HSiKAN and FIR are not synergistic (combining them gives no more than their sum).
- $\gamma \cdot C$ is dataset-dependent: $\gamma_{\text{Bitcoin}} = 0$ but $\gamma_{\text{Epinions}} > 0$ would say CPML helps only on hub-heavy graphs.

Statistical significance: paired t-test on the 5-seed deltas at each cell against its embedding-only floor. Effects with paired $|t| < 2$ are reported as "not distinguishable from noise."

3 The 8 cells (operational definition)

Each cell is a model class. Let $X^{(0)}$ be the learnable per-vertex embedding. Let $\mathrm{Agg}_{\mathrm{MLP}}$ denote the default MLP aggregator (always present as a fallback). Let $\mathrm{Agg}_{\mathrm{HSiKAN}}$ and $\mathrm{Agg}_{\mathrm{FIR}}$ be the optional aggregators of the corresponding axes. Let T denote a single forward through the topology (Flat = one layer of the active aggregators; CPML = L tier-stratified layers).

A	C	F	cell label	active components
0	0	0	`baseline`	MLP only, $L = 1$ (the embedding-floor reference)
0	0	1	`fir`	Clifford-FIR layer, $L = 1$
0	1	0	`cpml`	MLP layer per tier, $L = 3$
0	1	1	`cpml-fir`	Clifford-FIR per tier, $L = 3$
1	0	0	`hsikan`	Catmull-Rom signed-spline layer, $L = 1$
1	0	1	`hsikan-fir`	HSiKAN and Clifford-FIR in parallel, α-routed, $L = 1$
1	1	0	`cpml-hsikan`	HSiKAN per tier, $L = 3$
1	1	1	`cpml-hsikan-fir`	HSiKAN + FIR in parallel per tier, α-routed, $L = 3$

Routing when both A=1 and F=1: in the same layer, run HSiKAN and Clifford-FIR in parallel branches, then combine via a learnable softmax-normalised $\alpha = \mathrm{softmax}(\alpha_{\mathrm{logits}})$ of size 2:

$$\mathrm{Agg}_{A,F=1}(X) \;=\; \alpha_0 \cdot \mathrm{HSiKAN}(X) \;+\; \alpha_1 \cdot \text{Clifford-FIR}(X).$$

This is the same α-routing primitive that gave the Slashdot SOTA result via multi-arity routing — we re-use it here as inter-aggregator routing.

4 Affected files

New / modified (when reopened):

```
signedkan_wip/src/
        cpml.py
              ClifFIRTierAggregator              already shipped (today)
              HybridTierAggregator               NEW: hosts the  -routed
                                            HSiKAN + Clifford-FIR pair
              CPMLConfig (extended)              three boolean axes
        run_cpml_factorial_v2.py               NEW: 8-cell factorial
                                          runner with multi-seed +
                                          paired stats output
docs/plans/2026-05-11-hsikan-cpml-fir-orthogonal/
        plan.tex                               this doc
        plan.pdf                               built
        plan.tikz                              8-cell hyper-cube diagram
        plan.mmd                               Mermaid flow of the 3 axes
```

CORE.YAML items touched: none. All edits are within the existing CPML scaffolding and the factorial runner.

5 Test strategy

Pre-factorial unit tests:

- Each cell's forward + backward shape test (12 LOC each; parametric over (A, C, F)).

- No-cycles ablation for each cell (validates the embedding-floor measurement; if any cell's no-cycles AUC diverges from `baseline`'s no-cycles AUC, the floor estimation is biased and the cell's lift attribution is invalid).
- HybridTierAggregator α-routing: starts uniform (verify $\alpha_0 \approx \alpha_1 \approx 0.5$ at init); per-seed convergence trace tests confirm α drifts to non-uniform only when the data demands it.

Factorial measurement (per seed, per dataset, 5 seeds, 2 datasets):

- For each of the 8 cells, measure paired (cell, baseline) validation AUC at $n_{\text{epochs}} = 50$.
- Embedding-only floor measured once per (seed, dataset), used as the paired baseline for all 8 cells.
- Output: 5-seed paired + paired SD + paired t per cell per dataset. Plus the 7 factorial-decomposition coefficients $(\alpha, \gamma, \varphi, \alpha\gamma, \alpha\varphi, \gamma\varphi, \alpha\gamma\varphi)$ fitted on the 8-cell averages, with their estimated standard errors.

Validation gate: a single-axis effect (α, γ, or φ) is reported as "architecturally non-trivial" only if its 5-seed paired $|t| \geq 2$ on at least one dataset. An interaction ($\alpha\varphi$, etc.) is reported as "synergistic" only if its $|t| \geq 2$ AND both single-axis effects are non-trivial.

6 Performance budget

- Per-cell training time (Epinions, 50 epochs, warm cache): ~ 200 s.
- 8 cells $\times$ 5 seeds $\times$ 2 datasets = 80 runs.
- Wall: $\sim 4-6$ hours total (sequential, single CPU). GPU would compress to ~ 1 hour.
- Peak RSS per run: ≤ 4 GB (matches the existing `test_memory_below_4gb_at_scale` test in `test_cpml.py`).

7 Risk anticipation

- **Embedding-floor dominance**: today's prior factorial showed all aggregators converge to the embedding floor on random 80/20 Epinions. Mitigation: run on a structurally-harder split (e.g. time-stratified for Bitcoin OTC, or inductive-subset for Epinions) where the embedding-lookup advantage is weaker. Specify split protocol per dataset.
- **α-route degeneracy**: if HSiKAN and FIR branches collapse to one with $\alpha_0 = 1$ or $\alpha_1 = 1$, the hybrid cells are not meaningfully different from single-aggregator cells. Mitigation: report observed α distribution at end-of-training; if degenerate, the hybrid cells become hygienic checks rather than novel claims.
- **Statistical power**: 5 seeds with paired t-test can resolve effects $\geq\sim 1.5$ pp at $\alpha = 0.05$ when the per-seed SD is ~ 1 pp. Effects below this floor are not distinguishable from noise. Mitigation: report 95% confidence intervals on each coefficient; if intervals span zero, declare null result, not "no effect."
- **Cycle-cache fingerprint drift**: env-var-triggered cache invalidation already caught one silent bug today. Any new env var in the factorial (e.g. `HSIKAN_AGGREGATOR_KIND`) must be added to `cycle_cache._topk_fingerprint`. Already enforced by `feedback_cycle_cache_fingerprint.md`.

8 Rollback path

The 8-cell factorial is additive; no existing pipeline is modified. If results are uninformative, the factorial runner and the HybridTierAggregator can be ignored. The other architectural plans (Nagare, CPML, HymeKo-Nagare-to-HSMM) are independent.

9 Sequencing

1. Implement HybridTierAggregator + extend CPMLConfig (today's 1-hour task; trivial because Clifford-FIR is already a TierAggregator).
2. Unit tests for forward/backward shapes + α-routing init.
3. Single-seed dry run (8 cells $\times$ 1 seed $\times$ 2 datasets = 16 runs, $\sim$ 1 hour) to validate runner.
4. Full 5-seed sweep ($\sim 4-6$ hours) on the overnight queue.
5. Factorial decomposition + report.

10 Hypothesis register

Pre-registered hypotheses to test (with priors):

1. **H1 (FIR helps where HSiKAN doesn't)**: $\varphi > 0$ on Bitcoin OTC, $\alpha \approx 0$ on Bitcoin OTC. *Prior:* HSiKAN's spline expressiveness is wasted on small dense graphs; FIR's tiny parameter count regularises better.
2. **H2 (HSiKAN helps where FIR doesn't)**: $\alpha > 0$ on Slashdot/Epinions (large, heterogeneous), $\varphi \approx 0$. *Prior:* Catmull-Rom spline captures non-linear cycle importance that linear FIR misses.
3. **H3 (CPML helps multi-scale graphs only)**: $\gamma > 0$ on Epinions/WikiSigned (degree-skewed), $\gamma \approx 0$ on Bitcoin OTC (uniform). *Prior:* tier-stratified depth needs a multi-scale to exploit.
4. **H4 (HSiKAN + FIR synergistic on real-world graphs)**: $\alpha\varphi > 0$ on Slashdot, Epinions, WikiSigned. *Prior:* spline captures non-linearity, FIR captures sign-structure, they decompose orthogonally and compose constructively.

Pre-registration is the discipline that keeps the result honest; the bug-find pattern of the day suggests this is exactly the kind of work that needs it.

11 Venue

Primary: ICLR 2027 (clean architectural-ablation paper with formal factorial decomposition and pre-registered hypotheses).

Secondary: NeurIPS 2026 GLF workshop (small-paper version of the negative-result decompositions if the headline factorial fails to find a single-axis effect).

Tertiary: KDD applied (companion to the Slashdot-SOTA paper; uses the 3-axis factorial as the ablation section that justifies the chosen model architecture).